

Exam Date & Time: 10-Jan-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech V Semester - Make up Examination - Jan -2025

Cyber Security [IT-3156]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Identify and explain the different types of attackers in cybersecurity. Discuss how these attackers differ in their motivations and capabilities, and how they can exploit the different types of security vulnerabilities that exist in information systems (5)
 - A)
 - B) Consider the C-I-A triad and the four acts of harm are two fundamental concepts in information security. Analyse how these concepts are interrelated, and explain how the four acts can cause harm to the confidentiality, integrity, and availability of assets in a complex and interconnected system (3)
 - C) In the context of online services, explain how the concept of "identities are typically public or well-known, authentication should be private" can be applied to improve security. (2)
- 2) Biometric authentication systems are increasingly being used to verify identity and access control. However, these systems are not without their problems. Discuss the problems with biometric authentication systems and analyse their implications for security and privacy. (5)
 - A)
 - B) A large and complex organization has a decentralized access control system. Each department has its own access control policies and procedures, and there is no central authority responsible for coordinating access control across the organization.

Analyse the specific challenges that this organization would face in implementing a robust access control system? (3)

How would you recommend that this organization address these challenges?
 - C) Explain the key critical distinctions between AES and DES? (2)
- 3) Examine and elaborate on the multifaceted phases of ethical hacking. Provide a thorough analysis of each stage, highlighting the methodologies and ethical considerations involved. Clearly Explain the sequential progression through these phases and discuss their significance in the systematic identification and securing of potential vulnerabilities within information systems. (5)
 - A)
 - B) Critically analyse the challenges of developing and deploying effective security solutions to protect against viruses, worms, and trojans. Propose some potential

solutions to these challenges and discuss their feasibility and effectiveness.

- C) Elucidate the occurrence of buffer overflows, accompanied by a pertinent example that illustrates the mechanisms involved. (2)
- 4) Delve into a comprehensive elucidation detailing the various classifications of email attacks and critically analyse by providing a nuanced exploration of each type along with their distinct methodologies and potential ramifications. Subsequently, offer a thorough examination of viable solutions and countermeasures designed to mitigate the risks posed by these diverse email threats, ensuring a multifaceted understanding of the complex landscape of email security." (5)
- A)
- B) Explain a detailed depiction of a layered operating system, elucidating the sequential stages involved in the loading process of the operating system. Expound upon the intricate layers, unravelling the complexity inherent in their interplay, and subsequently, articulate the step-by-step progression through which the operating system is loaded in a systematic and staged manner. (3)

- C) **Analyse the following phishing email and identify the key elements that make it effective:**

Subject: Important security alert from [Bank Name]

Body:

Dear [Customer Name],

We have detected suspicious activity on your account and have temporarily suspended it.

To protect your account and funds, please click the following link to verify your identity: (2)

[Link to phishing website]

If you do not verify your identity within 24 hours, your account will be permanently closed.

Thank you for your cooperation.

Sincerely,

[Bank Name] Security Team

- 5) Cyberterrorism is considered by some defence analysts to be a larger threat than traditional terrorism. Justify this statement with a real-time example. What do you understand by Cyberterrorism? Explain the cyber threats which India is vulnerable to and bring out the state of the country's preparedness to deal with the same. (5)
- A)
- B) A company is developing a new facial recognition technology that can be used to identify and track individuals in real time. The company claims that the technology can be used to improve public safety by preventing crime and terrorism. However, there are also concerns about the potential for the technology to be misused to track and monitor individuals without their consent. (3)

Analyse the ethical and legal implications of the company's new facial recognition technology. Consider the following questions:

Explain potential benefits and risks of the technology?

How can the technology be used to improve public safety without violating individual privacy?

What laws and regulations should be in place to govern the use of facial recognition technology?

- C) Analyse the role of Risk Analysis in cybersecurity and delve into its complexities. Identify three key factors that contribute to the challenges of conducting a thorough cyber risk analysis. (2)

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